# Cloud and Fog based Integrated Environment for Load Balancing using Cuckoo Levy Distribution and Flower Pollination for Smart Homes

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***Abstract*—Reducing delay and latency in cloud computing environment is a challenging task for the research community. There are several smart cities in the world. These smart cities contain numerous Smart Communities (SCs), which have number of Smart Buildings (SBs) and Smart Homes (SHs). They require resources to process and store data in cloud. To overcome these challenges, another infrastructure fog computing environment is introduced, which plays an important role to enhance the efficiency of cloud. The Virtual Machines (VMs) are installed on fog server to whom consumers' requests are allocated. In this paper, the cloud and fog based integrated environment is proposed. To overcome the delay and latency issues of cloud and to enhance the performance of fog. When there are a large number of incoming requests on fog and cloud, load balancing is another major issue. This issue has also been resolved in this paper. The load balancing algorithm Cuckoo search with Levy Walk distribution (CLW) and Flower Pollination (FP) are proposed. The proposed algorithms are compared with existing Cuckoo Search (CS) and BAT algorithm. The comparative analysis of these proposed and existing techniques are performed on the basis of Closest Data Center (CDC), Optimize Response Time (ORT) and Reconfigure Dynamically with Load (RDL). The RT of DCs of cloud and clusters, Processing Time (PT) of fogs is also optimized on the basis of CLW and FP.**

***Index Terms*—Cloud Computing, Fog Computing, Smart Homes, Load Balancing, Service Broker Policy, Cuckoo search with Levy Distribution, Flower Pollination, Cuckoo Search, BAT**

## I. Introduction

The cloud computing is the very tremendous domain, where a large number of requests and intensive tasks are processed on demand basis. The cloud computing has some valuable features such as resource pooling and elasticity, self and on-demand services, pricing, quality of services and scalability. Due to which cloud based services are provisioned to consumers' at every level, where the resource management plays an important role. The resources and services provided to the consumers' exist in an industrial area or may exist in Smart Homes (SHs) residential area. When the requests are sent to the cloud from the consumer side of residential and industrial side, because of its remote location, delay arises. So, the Response Time (RT) and Processing Time (PT) of cloud Data Centers (DCs) are also increased. The other issue which occurs because of distant location is known as latency issue. To overcome the mentioned issues and overlapping the peculiarities of cloud, fog is recently introduced by Cisco.

The fog has additional attributes of location awareness and edge DC deployment because of its edge location. The distributed computing solutions, to achieve high performance in terms of scalability and elasticity is supported by fog computing. The Virtual Machine (VM) based environment is introduced in fog computing to reduce the computational cost and increases the sharing of information and resources.

The Internet of Things (IoT) gains popularity on a regular basis because in this era every object is connected over the internet. Because of this connectivity, most of the surroundings and objects become smarter and intelligent [1]. Now, it is easy to enhance the performance of objects and remove end to end delay. It also plays a vital role in convert traditional grid into Smart Grid (SG) equipped with some communication equipment.

When SG is introduced, it means that there is a two-way communication between the grid and consumer side. The city, buildings, homes located at consumer side converted into smart cities, Smart Buildings (SBs) and SHs. The SHs are modern ways of living. Owing to smart environment,

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the status of SHs consumption can easily be monitored by SG. When there is two-way communication between SBs and SG, the consumers' can easily reduce their electricity bill by selling their extra energy to SG or other SB. The cloud and fog also provide efficiency to the smart environment by handling the challenges of SBs and SHs. The cloud provides storage of extra energy of SHs, SBs and SCs. The fog provides nearer grids and resources to the consumers' to run their applications and requests [2].

The cloud and edge cloud plays an impactful role in smart mobiles [3]. The processed state synchronization is introduced to reduce the execution time of the cloud. The RT and connectivity issue is also resolved because of edge cloud. To reduce the cost and to play a vital role in terms of low power computing in [4], cloud edge is used. The cloud edge also reduces the power consumption and execution time of DCs. In [5], to enhance the features of cloud and edge cloud. The purpose of introducing this fog is to minimize the latency and delay in the cloud based environment.

The Chaotic Social Spider Algorithm (CSSA) is proposed to make a solution for efficient utilization resources. The overall makespan of cloud computing is reduced and load balancing is done on the DCs of the cloud [6]. The Stackelberg is proposed in cloud computing environment [7]. The purpose to implement this game theory approach to minimize the cost and find appropriate information of resource allocation of DCs of cloud.

### *A. Contributions*

In the summary of previous studies and researches, these things revealed that how cloud give their best performance in IoT. How cloud and fog computing enhance the efficiency of SG and how it converts homes into SHs. Keeping in mind the previous studies, in this paper, we implemented cloud and fog based integrated environment. In the direction of efficiency and performance of implementing the system, the contribution of this paper could be summarized below:

- Cloud and fog base integrated environment is implemented for six regions of the world,
- Cloud of this paper gives the storage facility to the consumers' of SHs,
- Fogs of implementing scenario overcome the delay and latency issue,
- Load balancing is performed, to balance the load of fog and cloud,
- Cuckoo search by using Levy Walk distribution (CLW) and Flower Pollination (FP) is proposed for load balancing,
- Comparison of the proposed techniques and existing techniques Cuckoo Search (CS) and BAT algorithms are also performed,
- The comparison of techniques is performed on the basis of Closest Data Center (CDC), Optimize Response Time (ORT) and Reconfigure Dynamically with Load (RDL),
- RT of the proposed techniques are minimized,
- PT of proposed CLW and FP is also decreased on ORT and RDL service broker policy,
- VM cost, Micro Grid (MG) cost, Data Transfer (DT) cost and Total Cost (TC) of proposed CLW and FP are also minimized.

The remainder of the paper is organized as follows: related studies are presented in Section II. system model with the proposed methodology is demonstrated in Section IV. Section V describes the simulation results of our proposed schemes. The section VI describes the conclusion of this paper.

## II. Related Work

The fog computing is proposed as the extension of cloud computing in [8]. The fog provides computational and storage services to the SG. This also deals with the fault tolerance and line loss problem of the SG. The task scheduling algorithm is proposed because it is also necessary to complete all tasks on time. The aim of this proposed mechanism is to schedule the tasks on time and to reduce the delay and latency between the fog nodes. Delay is reduced on the basis of reallocation mechanism. However, to make the system simpler the resources and computational time is omitted. Dynamic Clustering League Championship Algorithm (DCLCA) is proposed in [9]. The purpose is to deal with the fault tolerance problem. In previous researches, the schemes and algorithms are proposed for scheduling. In this work, they considered ignoring the fault tolerance problem by using an implemented algorithm. The aim of this proposed work is to deal with task allocation with fault tolerance awareness problem in cloud computing. On the other hand, the performance of this proposed algorithm on the implemented system is not discussed.

Carrasco *et al.* presented new standard Topology and Orchestration Specification for Cloud Applications (TOSCA) and CAMP to satisfy the consumer level. It is because they provide services to the customer on the basis of three platforms, i.e., Infrastructure as a Service (IaaS), Platform as a Service (PaaS) and Software as a Service (SaaS) to meet the Quality of Services (QoS) and Service Level Agreement (SLA). In this research work, the TOSCA based model is proposed for Paas and Saas to rigorously followed. CAMP concepts are proposed to build unified Application Programming Interface (API). The proposed works satisfy the services on PaaS as compared to SaaS [10]. The server selection, configuration, reconfiguration and automatic performance technology is proposed by Yamato in [11]. The aim of this proposed technology is also to meet the service level of cloud on different levels. Because of various level of cloud, there is also the number of servers such as CPU servers, container and baremetal server. To differentiate the proposed technologies with

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previous researches, named it as Field Programmable Gate Arrays (FPGAs). The implemented technology works on the basis of GPUs servers. The implemented technology proves VMs servers performs better than a CPU server. However, it is complex to implement and understand for simple scenario.

In [12], multi-tier fog computing with large scale data analytics is proposed for smart cities applications. The aim of this proposed model is to tackle the problem of computing infrastructure and less responsiveness of cloud. The proposed model is for large-scale data handling and to meet the QoS for the consumers'. This model also contains ad hoc nodes to processes the incoming requests. The resources present inside this model are shared resources, due to which the performance of this model increases to meet the QoS scheme. However, other attributes of this proposed model, e.g. implemented cost, resource utilization cost, RT and PT of proposed model can be discussed. In [13], author's discussed the problem of completion time of cloud. Because it is really challenging tasks to schedule the applications when there is a large number of Scientific Work Flow Applications (SWFA). These SWFA effects the performance and optimization cost of cloud, due to various number of VMs and size of DCs. The Completion Time Driven Hyper-Heuristic (CTDHH) on the basis of Low Level Heuristic (LLH) to optimize the cost and reduces the completion time of cloud resources. The implemented LLH compared with Hyper-Heuristic Scheduling Algorithms (HHSA) that are Genetic Algorithm (GA), PSO, etc. After implementation CTDHH performs better and less computational time than other algorithms. On the other hand, they consider load balancing and fog base integrated model in their work to achieve more efficiency.

The Water Constrained workload scheduling (WATCH) for DCs is proposed in [14]. In this research, by implementing this WATCH algorithm, they reduced the workload on cloud DCs and also aimed to minimize operational cost. The proposed scheduling algorithm is proposed keeping in mind the whole geographic location of the world. This proposed WATCH follows the Geographic Load Balancing (GLB). This proposed algorithm follows the attributes of water, where DCs seems free and have fewer requests when maximum size requests move towards other DC. The proposed algorithm also achieves the minimal operational cost. On the other side, they can consider RT and PT of the proposed scheduling algorithm.

Sotiriadis *et al.* proposed cloud VM scheduling algorithm in [15]. The proposed algorithm is to optimize the performance of VMs in an efficient manner. The aim of this work is to minimize the workload and degradation of workload on cloud DCs. When there is maximum number of requests on any VM and CPU of that utilize maximum resource, then it requests and shares the CPU of other VM to degrade the load of its DC. In this work, they also provided a comparative analysis of their proposed work with other traditional schedulers by studying the behavior of NoSQL (MongoDB, Apache Cassandra, and Elastic Search). After, they achieved minimization of load, increased CPU PT and decrease load on VMs because of an overload of PMs. On the other hand, they can introduce fog in their work, because of fog, the requests with minimum size can be processed on fog and load on the VMs of cloud automatically decreases.

## III. Problem Statement

The challenges and issues of delay, task scheduling, and distribution, load balancing, etc., arise because of the large sharing of web resources over the internet. The aforementioned challenges and their proposed solutions are described below.

The CSSA, inspired by social spider algorithm is proposed to tackle the problem of task scheduling in [6]. In [16], to solve the delay optimal VM scheduling problem in cloud, Shortest Job First (SJF) combines with Min-Min Best Fit (MMBF) and proposed new scheduling scheme that is SJF-MMBF to solve the delay problem of cloud. The SJF is also combined with Reinforcement Learning (RL) to solve the starvation issue of the requests. However, in [6] and [16] the RT and PT of proposed algorithm are not calculated.

The Energy Aware Load Balance Scheduling (ELBS) is proposed in [17] on fog based platform. The purpose of this proposed scheduler is to solve the complex energy consumption problem of clusters and balance the load. To make this solution more optimal, PSO is proposed. The performance of the proposed scheduler is improved, because of an optimal solution. However, the processing time of the scheduler and swarm optimization technique is not calculated. The tabu search method for load balancing in fog environment is proposed in [18]. The computational cost of processing tasks in a fog based environment is reduced. However, RT and implementation cost of VMs are not calculated. To solve bi-objective minimization problem, metaheuristic techniques PSO, Binary PSO (BPSO) and BAT algorithm are proposed in [19]. The allocation of requests to VMs on fog based environment is considered as bi-objective minimization problem in this research. The BAT performs as compared to other two algorithms and tasks are efficiently assign to VMs. The energy is also minimized in their proposed scenario. However, load balancing is not considered.

The concept of Fog of Things (FoT) is introduced in [20] to overcome the challenges of responsiveness, scalability, heterogeneity and handling of large and complex data, etc. in the IoT. The robustness and location-aware of the things are achieved. Still, scalability issue is not resolved. On the other hand, cloud can be considered in this work to solve the scalability issue.

## IV. Proposed System Model

The goal of this proposed work is to minimize the latency and delay in cloud and fog based emerging platform.

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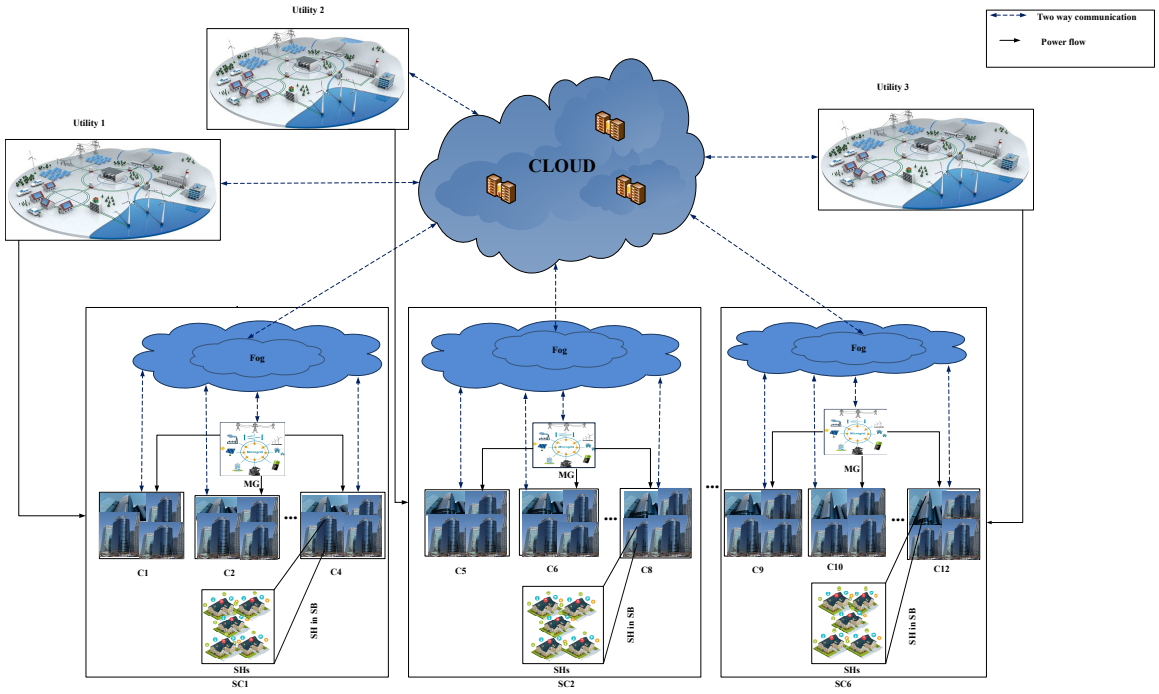



Fig. 1: Smart Environment on the Basis of Cloud and Fog integrated Environment

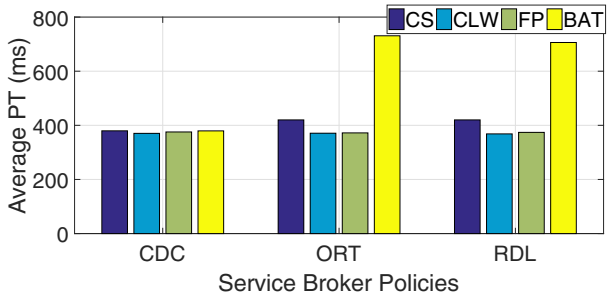



Fig. 2: Average PT

This proposed work is aimed to utilize the resources in an efficient manner. To justify the work of the proposed model shown in Fig. 2, the comfort to the consumers' at every step of this smart environment is provided. This proposed model consists of six number of Smart Communities (SCs). Each community has the number of clusters and each cluster has its own fog. Every community has also its own utility that provides services against consumer requests. The utility1 provides the services to the SC1. The utility2 provides the services to the SC2 and the utility3 provides the services to SC3. The functionality of this architecture utilities is defined further in other layers.

### *A. Cloud Computing Layer*

The proposed cloud computing layer of this research work is described here. This cloud is implemented on the top of the SCs. It provides the same utilities and facilities as another cloud provides. To fulfill the consumer demand and needs on hourly bases, there is the number of DCs allocated inside the cloud. The allocated DCs processed the incoming data from the consumer side. These DCs check the status of incoming requests then process them according to the status of the request. If consumers' need electricity, the cloud processed on the request then this request is forwarded to the utility of that SC. There is two-way communication between cloud and utility. The required electricity is sent to the consumers' and remaining is kept stored inside the backup storage of the cloud. The record of SBs, SHs, and SCs are also updated on the cloud. SCs cloud maintains the record of every consumer on daily basis. The purpose of keeping the record of the consumers' is that the utility can easily get information of the regular users. When the request of consumer forwarded to the utility at cloud layer. The utility provides some knowledge which is based on the current electricity rate. The cloud forwards that requests to the consumers' through a fog. The utility of other five communities and cloud performs the same functionality.

### *B. Fog Layer*

The fog layer enhances the performance of the cloud. As in previous researches [21], this thing describes that fog is located at the edge of the network and reduces the latency and delay of the cloud because of its two-fold quality. The second one is that the consumer requests processed faster or earlier at the edge of the network. In this research study, proposed fog layer performs the same functionality as done in previous researches. This proposed fog computing layer is integrated with the cloud

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to improve the efficiency and credibility of the cloud. The several PMs and VMs are located at the fog edge which executes and process the consumers' requests. In this study, fog also provides the two way communication between cloud and consumer layer. Each cluster has its own fog that provides the location services to that cluster. Every community has two number of fogs that makes the process fast and provides maximum response to the consumer request. When the consumer request comes to the fog the described functionality of described hardware resources provides that services to that request. After the execution of the request, if the consumer demand can be fulfilled on fog level then this request forwarded to MG. If the demand of the consumer is high or need more electric power than fog forwards this request to cloud. Every fog of each six communities provides the same services.

*1) Micro Grid:* MG consist of renewable energy resources and each community has 2-5 its own MG. The MG is located between the SBs and Fog. In this proposed study, MG and fog both can communicate with each other. When the requests from fog come to the MG, it gives the response back to the fog that how much power consumption or for how many homes it can provide services. The power is supplied from MG to SBs and SHs of this proposed study.

### C. Consumer Layer

By combining SHs, SBs and SAs the SC of this proposed environment is generated. The bottom-most layer of this research is the consumer layer and which presents the smart environment of this proposed work. In this implemented work, SBs are located inside the consumer layer, each SB has number have SHs also presented in this proposed model. The SHs are located, these are SHs because of smart meters are attached to them. They can share their extra energy to other SHs and with the utility by communicating with fog and cloud. The SAs that are located inside the SHs also make them smart. Due to automatic control system these appliances are called SAs, because they can easily turn on or turn off from the central controller during on and off-peak hours. The controllers are located on fog and cloud, and easy to operate these SAs directly. There is also two fold functionality of this layer, the first one is, the requests for services from this layer is to the upper layers of the proposed model. Second is that, when the request is processed, this layer get feedback from those layers and use the requested power. The functionality of each cluster, each SHs and each consumer is the same in every community.

## V. Simulation Results and Discussion

To evaluate the performance of our proposed techniques and existing heuristic techniques, cloud analyst is used. The performance evaluated via several parameters such as RT, PT and cost of the proposed and existing techniques such as CS, BAT and proposed CLW and FP. Moreover,

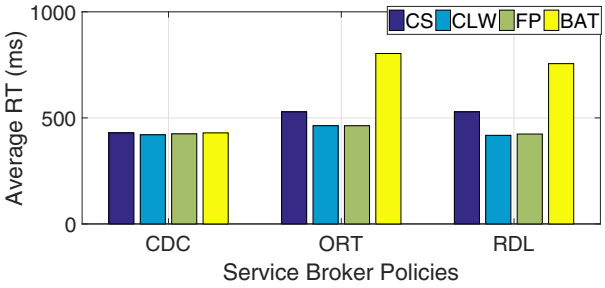



Fig. 3: Average RT

this paper has implemented some scenarios. The scenario used for evaluation of this paper consists of six regions of the world and each region contains two clusters and one fog is also attached with these clusters. To process the requests of the consumers' of the clusters, 2-5 VMs are allocated to these clusters. The 5-10 MG are used to fulfill the demands of these SHs exists in these clusters. There are total 12 number of clusters in this proposed scenario and each cluster of SHs have 10000 number of requests. The results of proposed scenario and techniques which are used for load balancing are compared on the basis of CDC, ORT and RDL service broker policies of the cloud analyst.

### A. PT

The Fig. 2, shows the comparative analysis of proposed and existing techniques on the basis of PT. The results show that the PT of FP and CLW is less than other existing techniques. It defines that, the proposed schemes of this paper perform better and process large number of consumers' requests on less delay and latency. A large number of consumers' requests processes and executes on the basis of mentioned three policies. BAT takes maximum time to process the consumers' requests on the basis of ORT and RDL policies.

### B. RT

The RT of these load balancing techniques are compared in Fig. 3. The results revealed that the FP and CLW have less RT on the basis of ORT and RDL. The numerous requests of consumers' processed and responded with less latency. On these two policies, CS and BAT is not performed better.

The RT of CLW on CDC is 32.40%, ORT is 35.69% and RDL is 31.89%. It means that on RDL policy CLW has better results.

### C. Cost

In Fig. 4 shows the cost of load balancing algorithms. The VM cost, MG cost, DT and TC are compared. The comparative analysis revealed that CS, CLW and BAT has less VM cost except for BAT algorithm. The VM cost of these mentioned load balancing algorithms are less due to 2-5 fix number of VMs. The MG cost of these algorithms are also minimum, it is because these algorithms balance

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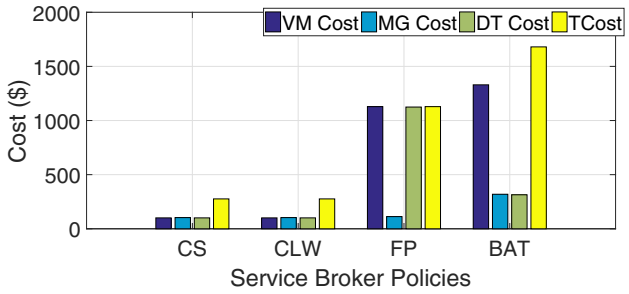



Fig. 4: Cost Comparison

the load on fog and cloud. So, that's why there is no more need for MG to fulfill the energy requirements. The DT cost of CS, FP and BAT are high except CLW. This Fig. 4 shows that the proposed techniques of this paper CLW and FP have less total cost as compared to CS and BAT. Overall, BAT consumes maximum cost as compared to other three techniques.

## VI. Conclusion

This paper proposes nature inspired load balancing algorithms, CLW and FP in cloud and fog based integrated environment. The scenario is modeled to access the effects of proposed model and load balancing techniques. The effects of the proposed model are evaluated by utilizing the cloud analyst tool. The primary intent of proposed techniques on cloud and fog environment is to minimize RT, PT and cost. Our experiment results show that the FP and CLW have minimum RT and PT as compared to other two techniques. These results are taken on the basis of CDC, ORT and RDL service broker policies. The section V shows the detailed results of our proposed techniques. The cost is also decreased by using our proposed algorithms FP and CLW. The Future work involves looking into the problem of VM migration and bin packing problem. The individual task processing time will also be considered.

## References

[1] Li, Shichao, Ning Zhang, Siyu Lin, Linghe Kong, Ajay Katangur, Muhammad Khurram Khan, Minming Ni and Gang Zhu. "Joint admission control and resource allocation in edge computing for Internet of things." IEEE Network 32, no. 1 (2018): 72-79.

[2] Yassine, Abdulsalam, Shailendra Singh, Shamim Hossain and Ghulam Muhammad. "IoT big data analytics for smart homes with fog and cloud computing." Future Generation Computer Systems 91 (2019): 563-573.

[3] Ahmed, Ejaz, Anjum Naveed, Abdullah Gani, Siti Hafizah Ab Hamid, Muhammad Imran and Mohsen Guizani. "Process state synchronization-based application execution management for mobile edge/cloud computing." Future Generation Computer Systems 91 (2019): 579-589.

[4] Verba, Nandor, Kuo-Ming Chao, Jacek Lewandowski, Nazaraf Shah, Anne James and Feng Tian. "Modeling industry 4.0 based fog computing environments for application analysis and deployment." Future Generation Computer Systems 91 (2019): 48-60.

[5] Agostino, Daniele, Lucia Morganti, Elena Corni, Daniele Cesini and Ivan Merelli. "Combining Edge and Cloud computing for low-power, cost-effective metagenomics analysis." Future Generation Computer Systems 90 (2019): 79-85.

[6] Xavier, VM Arul and Annadurai. "Chaotic social spider algorithm for load balance aware task scheduling in cloud computing." Cluster Computing (2018): 1-11.

[7] Wei, Xunli Fan, Houbing Song, Xiumei Fan and Jiachen Yang. "Imperfect information dynamic stackelberg game based resource allocation using hidden Markov for cloud computing."IEEE Transactions on Services Computing 11, no. 1 (2018): 78-89.

[8] Yin, Luxiu, Juan Luo and Haibo Luo. "Tasks s andcheduling and resource allocation in fog computing based on containers for smart manufacture." IEEE Transactions on Industrial Informatics (2018): 4712-4721.

[9] Latiff, Muhammad Shafie Abd, Syed Hamid Hussain Madni and Mohammed Abdullahi. "Fault tolerance aware scheduling technique for cloud computing environment using dynamic clustering algorithm." Neural Computing and Applications 29, no. 1 (2018): 279-293.

[10] Carrasco, Jose, Francisco Durán Ernesto Pimentel. "Trans-cloud: CAMP/TOSCA-based bidimensional cross-cloud." Computer Standards & Interfaces 58 (2018): 167-179.

[11] Yamato, Yoji. "Server selection, configuration and reconfiguration technology for IaaS cloud with multiple server types."Journal of Network and Systems Management 26, no. 2 (2018): 339-360.

[12] He, Jianhua, Jian Wei, Kai Chen, Zuoyin Tang, Yi Zhou and Yan Zhang. "Multi-tier fog computing with large-scale IoT data analytics for smart cities." IEEE Internet Things J. (2017):1-10.

[13] Alkhanak, Ehab Nabiel and Sai Peck Lee. "A hyper-heuristic cost optimisation approach for Scientific Workflow Scheduling in cloud computing." Future Generation Computer Systems (2018):480-506.

[14] Islam, Mohammad, Shaolei Ren, Gang Quan, Muhammad Shakir and Athanasios Vasilakos. "Water-constrained geographic load balancing in data centers." IEEE Transactions on Cloud Computing 5, no. 2 (2017): 208-220.

[15] Sotiriadis, Stelios, Nik Bessis and Rajkumar Buyya. "Self managed virtual machine scheduling in Cloud systems. Information Sciences 433 (2018): 381-400.

[16] Guo, Mian, Quansheng Guan and Wende Ke. "Optimal Scheduling of VMs in Queueing Cloud Computing Systems With a Heterogeneous Workload." IEEE Access 6 (2018): 15178-15191.

[17] Wan, Jiafu, Baotong Chen, Shiyong Wang, Min Xia, Di Li and Chengliang Liu. "Fog Computing for Energy-aware Load Balancing and Scheduling in Smart Factory." IEEE Transactions on Industrial Informatics (2018): 4548-4556.

[18] Téllez, Nadim, Miguel Jimeno, Augusto Salazar and Nino-Ruiz. "A Tabu Search Method for Load Balancing in Fog Computing." Int.Artif. Intell 16, no. 2 (2018):1-31.

[19] Mishra, Sambit Kumar, Deepak Putha, Joel JPC Rodrigues, Bibhudatta Sahoo and Eryk Dutkiewicz. "Sustainable Service Allocation using Metaheuristic Technique in Fog Server for Industrial Applications." IEEE Transactions on Industrial Informatics (2018): 4497-4506.

[20] Yu, Ruozhou, Guoliang Xue, Vishnu Teja Kilari and Xiang Zhang. "The Fog of Things Paradigm: Road toward On-Demand Internet of Things." IEEE Communications Magazine56, no. 9 (2018): 48-54.

[21] Pan, Jianli and James McElhannon. "Future edge cloud and edge computing for internet of things applications." IEEE Internet of Things Journal 5, no. 1 (2018): 439-449.

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